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Autoscaling in Kubernetes

Instructor

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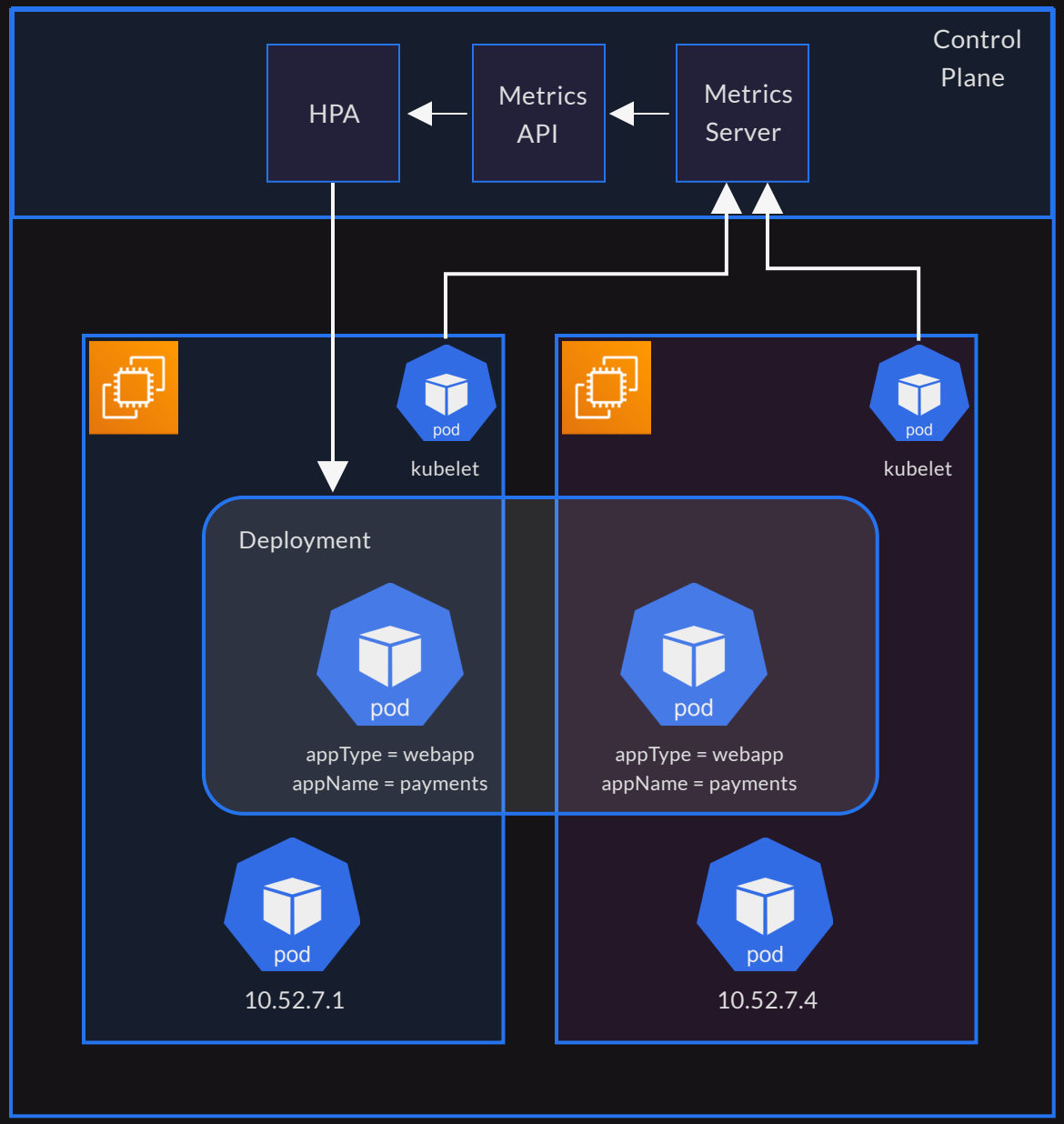
Autoscaling

Challenges in scaling:

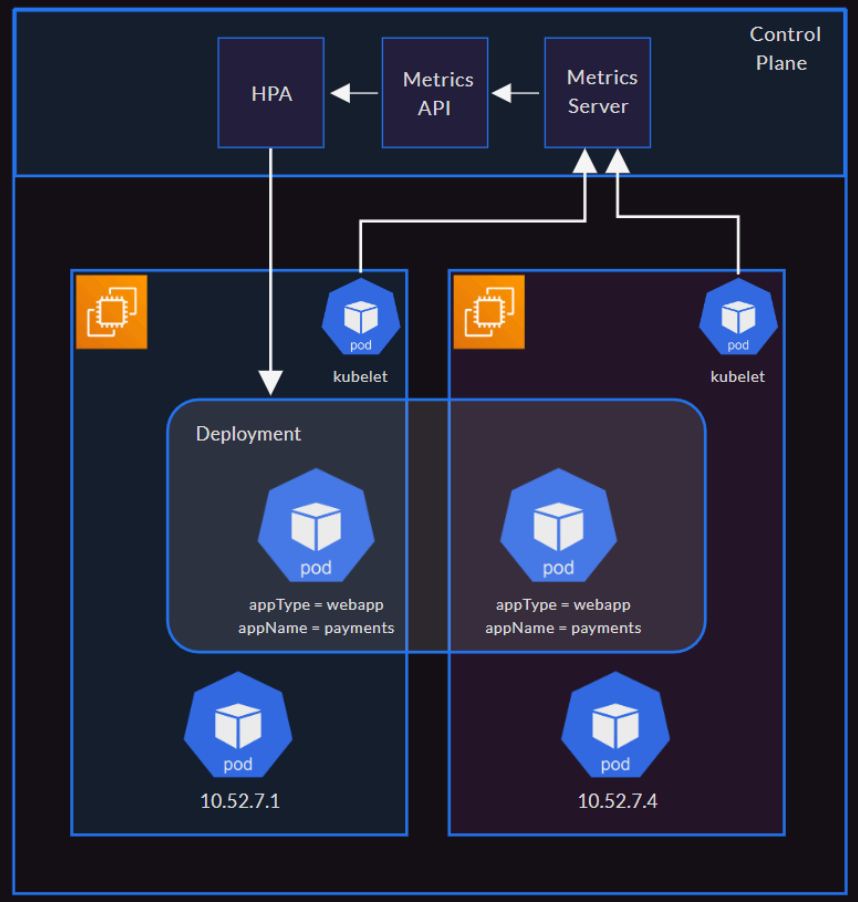
- Automatically scale containers based on certain metrices like CPU, memory, RPS (request per second)
- New containers to be automatically created and assigned to nodes
- New nodes to be automatically created with all the necessary OS and packages
- Manage nodes and container's health
- Remove nodes when not needed to save cost
- Graceful termination of containers to avoid bad user experience

HPA (Horizontal Pod Autoscaler)



HPA (Horizontal Pod Autoscaler)

- Automatically updates a workload resource such as a Deployment, to automatically scale the workload to match the demand
- Vertical scaling would mean assigning more resources (for example: memory or CPU) to the Pods that are already running for the workload
- There is also a cluster autoscaler that scales nodes based on the usage metrics



Project Details

- See manual scaling in deployments
- Setup HPA based autoscaling

Thank You